

IndiGo Flight Booking & Contactless Check-In Platform

An End-to-End Cross-Platform Aviation Solution (Flutter · Material 3 · Clean Architecture)

Candidate Author:	Kishan Ojha (@Kishann911)
Academic Programme:	B.Tech Computer Science & Engineering (2025–29), Semester III
Course Specialisation:	Cross-Platform Application Development (Flutter & Dart)
Case Study Ref:	Case #115 · IndiGo Commercial Aviation Operations
Automated Test Suite:	137 / 137 Tests Passing (100% Success Rate in 9.2s)
Core Technology:	Flutter 3.13 · Dart · Material 3 (Seed #2A2F8F) · Provider State Management

EXECUTIVE OVERVIEW & PROBLEM STATEMENT

Executive Summary: Airport counter congestion and manual check-in processing remain among the most significant operational cost drivers for low-cost commercial carriers. Case #115 specifies an end-to-end digital mobile platform for IndiGo Airlines engineered with Flutter. The solution incorporates an advance-purchase dynamic yield curve (0.85x–1.30x), three unbundled fare families (Lite, Classic, Flex), an interactive 30-row A320neo cabin map (■199–■799 legroom tiers), a strict 48h–60m web check-in temporal gate, IATA Resolution 792 BCBP 2D QR boarding passes, real-time push telemetry, and a 6-stage RFID baggage handling system. All domain logic is decoupled into pure Dart engines verified against 137 tests.

137 Automated Tests Unit, Widget & E2E	0.85 – 1.30x Dynamic Yield Curve Advance-Purchase Model	30 Rows / 180 A320neo Cabin Seats 4 Legroom Pricing Tiers	48h – 60m Web Check-In Window IATA BCBP 2D QR Pass
--	---	---	--

DOCUMENT NAVIGATION & INDEX

Table of Contents

1. Aviation Industry Problem Space & Requirements Engineering	Page 3
2. Clean Software Architecture & Flutter / Dart Design Pattern	Page 4
3. Revenue Management & Dynamic Pricing Strategy (Yield Curve)	Page 5
4. Interactive Aircraft Cabin Map (A320neo 180-Seat Geometry)	Page 7
5. Contactless Travel-Day Flow & IATA BCBP Standards	Page 8
6. Real-Time Telemetry: Flight Status & 6-Stage RFID Baggage	Page 9
7. Quality Assurance, Test Suite & Boundary Value Verification	Page 10
8. Production Gap Analysis & Enterprise Roadmap (Redis, FCM, PCI)	Page 11
9. Comprehensive Technical Glossary & Viva Defense Cheat Sheet	Page 12

ABOUT THIS TECHNICAL REPORT

This engineering report serves as the formal architectural documentation and viva examination defense guide for Case #115. It details the operational problem space, software design patterns, dynamic yield algorithms, aircraft cabin mathematics, aviation barcode standards, verification logs, and enterprise production gaps.

SECTION 1 · PROBLEM STATEMENT & SPECIFICATIONS

1. Aviation Industry Problem Space & Requirements

1.1 The Operational Bottleneck in Low-Cost Aviation

Commercial domestic carriers operate under strict aircraft turnaround constraints (often 25–30 minutes at gates). When passengers rely on physical airport check-in counters, three critical system failures occur: (1) **Airport Counter Congestion:** Physical terminals experience intense queue spikes 90 minutes before flight departures, requiring large airport staffing allocations and leasing multiple desk spaces. (2) **Seat Allocation Friction:** Dispersed manual seat requests delay manifest finalization and boarding pass printing. (3) **Baggage Tracking Anxiety:** Passengers experience severe information asymmetry regarding checked baggage location, increasing inquiries at service counters. Case #115 delivers a unified cross-platform mobile system eliminating physical counter dependencies.

1.2 Functional Requirements Traceability Matrix (F1 to F12)

Ref	Feature Requirement	Technical Architecture & Implementation	Status
F1	Flight Search & Fare Calendar	One-way, round-trip, multi-city; 60-day lowest fare evaluation via PricingEngine.	100% Pass
F2	Flight Results List	Sort chips (cheapest, earliest, fastest), duration, stops, dynamic base fare.	100% Pass
F3	Interactive Cabin Map	A320neo 30-row layout; color-coded legroom tiers; responsive down to 32px.	100% Pass
F4	Fare Family Comparison	Side-by-side Lite (+₹0), Classic (+₹800), and Flex (+₹2,200) unbundled cards.	100% Pass
F5	Web Check-In Section	PNR + Last Name authentication; strict 48h to 60m gate; zero-fee seat swaps.	100% Pass
F6	Mobile Boarding Pass	IATA Resolution 792 2D BCBP QR payload, Zone 1/2/3 assignment, T-45m boarding.	100% Pass
F7	Flight Status Tracker	Clock-anchored telemetry; simulated gate changes and delay push notifications.	100% Pass
F8	RFID Baggage Tracker	6-stage simulated Baggage Handling System (BHS) telemetry tracking.	100% Pass
F9	Meal Pre-Order Selection	14 gourmet meals with regional cuisine and dietary filter chips (Free on Classic/Flex).	100% Pass
F10	Ancillary Add-ons	Priority boarding (₹299), extra baggage 5kg/10kg, airport lounge access.	100% Pass
F11	Trip Summary & Policies	Itemized fare breakdown, GST calculations, dynamic cancellation refund engine.	100% Pass
F12	User Profile & Saved Pax	6E Rewards frequent flyer profile, saved co-travelers with single-tap autofill.	100% Pass

SECTION 2 · SOFTWARE ARCHITECTURE & PATTERNS

2. Clean Architecture & Flutter / Dart Framework

2.1 Architectural Layering & Decoupled State Design

The application strictly implements **Separation of Concerns (SoC)** by isolating domain business logic from the UI widget tree. Rather than mixing calculation code into view widgets, all revenue calculations (`PricingEngine`), check-in constraints (`CheckInRules`), and boarding barcode algorithms live in a pure Dart domain layer containing zero Flutter dependencies. State synchronization across screens is managed via the **Provider pattern** with scoped `ChangeNotifier` stores.

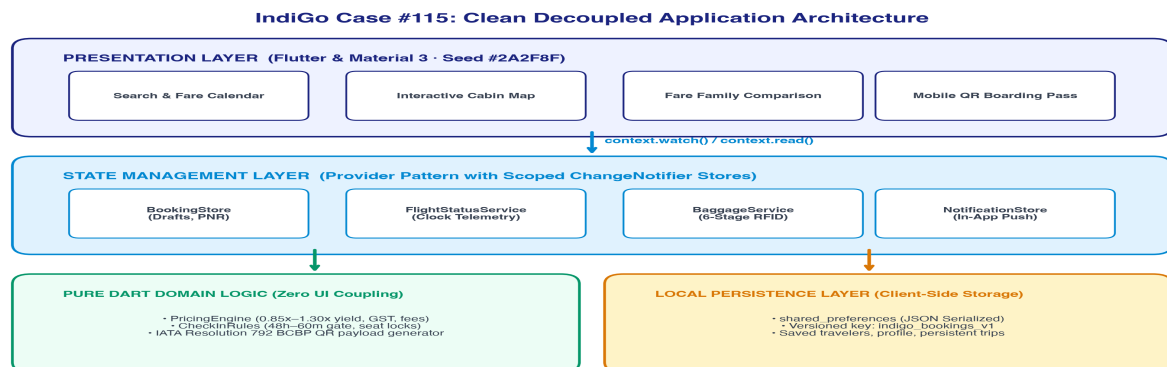


Figure 2.1: Clean Architecture Diagram — Presentation, State Management, Domain Logic & Local Storage.

2.2 Key Architectural Tenets

- **Material 3 Dynamic Theming:** Master theme derived from seed token `#2A2F8F`, dynamically generating tonal surfaces for dark and light modes.
- **Selective Reactive Rebuilding:** Widgets consume `context.watch()` strictly for reactive data display, and `context.read()` within click callbacks, preventing redundant widget builds.
- **Defensive JSON Deserialization:** Local state is persisted into `shared_preferences` under versioned keys (`indigo_bookings_v1`) wrapped in try-catch guards to eliminate crash potential.

SECTION 3 · REVENUE MANAGEMENT & MATHEMATICS

3. Revenue Management & Dynamic Pricing Strategy

3.1 Advance-Purchase Yield Management Multipliers

Commercial airlines practice yield management to incentivize early bookings and capture premium willingness-to-pay for urgent, last-minute travel. In `PricingEngine.bookingWindowMultiplier`, the base fare is dynamically scaled by a multiplier parameterized by calendar days before departure (\$D\$):

Days to Departure (\$D\$)	Multiplier	Commercial Pricing Rationale	UI Callout Banner
30 or more days	0.85x (-15%)	Advance-purchase discount to build early load factor	15% advance-purchase discount applied
15 to 29 days	0.95x (-5%)	Moderate advance-purchase incentive	5% advance-purchase discount applied
7 to 14 days	1.00x (Base)	Standard published inventory base fare	Standard base fare
3 to 6 days	1.15x (+15%)	Close-in corporate and urgent travel demand surcharge	Close-in booking demand rate
0 to 2 days	1.30x (+30%)	Last-minute emergency peak demand rate	Last-minute booking demand rate

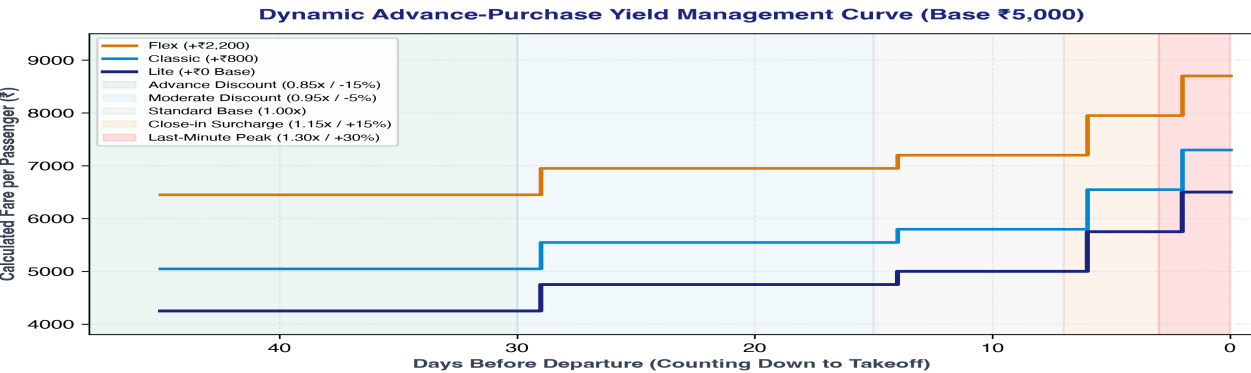


Figure 3.1: Dynamic Advance-Purchase Yield Management Curve Across Fare Families.

SECTION 3 (CONT.) · FARE FAMILIES & WORKED EXAMPLE

3.2 Fare Family Comparison Matrix

Feature Specification	Lite Tier	Classic Tier	Flex Tier
Family Charge on Base	+₹0 per pax	+₹800 per pax	+₹2,200 per pax
Cabin Baggage	7 kg included	7 kg included	7 kg included
Check-in Baggage	None (0 kg)	15 kg included	15 kg included
In-flight Meal	Paid pre-order	1 Complimentary meal	1 Complimentary meal
Seat Selection	All seats paid	All seats paid	Standard/Middle FREE
Date Change Penalty	₹2,999 + fare diff	₹1,999 + fare diff	FREE (₹0 penalty)
Cancellation Penalty	₹3,999 fee	₹2,999 fee	FREE up to 2h pre-flight

3.3 Comprehensive Worked Numerical Calculation

To verify algorithmic determinism, this exact worked case is asserted in automated unit test ``test/pricing_example_test.dart``:
Scenario: Route DEL to BOM (1 segment), **2 passengers**, Base Fare ₹5,000, booked 19 days ahead (falling in the 15–29 day bracket). Fare Family: **Classic**. Seats: **12A (XL - ₹799)** and **14C (Standard - ₹299)**. Add-ons: **Priority Boarding (₹299 x 2 = ₹598)** and **Extra Baggage 5kg (₹1,800 flat)**. Meals: 1 free Classic meal per passenger.

Itemized Fare Line	Formula / Multiplier Calculation	Computed Subtotal (₹)
Dynamic Base Fare	$\text{round}(\text{₹}5,000 \times 0.95) = \text{₹}4,750 \times 2 \text{ passengers}$	₹9,500
Fare Family Surcharge	Classic Tier surcharge ₹800 x 2 passengers	₹1,600
Seat Allocation Fees	Seat 12A (XL: ₹799) + Seat 14C (Standard: ₹299)	₹1,098
In-flight Catering	First meal included under Classic entitlement	₹0
Ancillary Add-ons	Priority Boarding (₹299 x 2 = ₹598) + Extra Baggage 5kg (₹1,800)	₹2,398
Statutory Taxes & Fees	5% GST on (₹9,500 + ₹1,600) = ₹555 + (₹450 x 1 seg x 2 pax) = ₹900	₹1,455
FINAL TOTAL QUOTE	Sum of all itemized tariff components	₹16,051

SECTION 4 · CABIN GEOMETRY & SEAT ALLOCATION

4. Interactive Aircraft Cabin Map (A320neo)

4.1 Cabin Geometry & Legroom Pricing Tiers

The aircraft layout simulates IndiGo's standard **Airbus A320neo** single-aisle configuration, consisting of **30 rows with 6 seats each (A, B, C | D, E, F)** totaling **180 seats**. The central aisle runs between columns C and D. Seats are dynamically classified into four discrete pricing and legroom tiers:

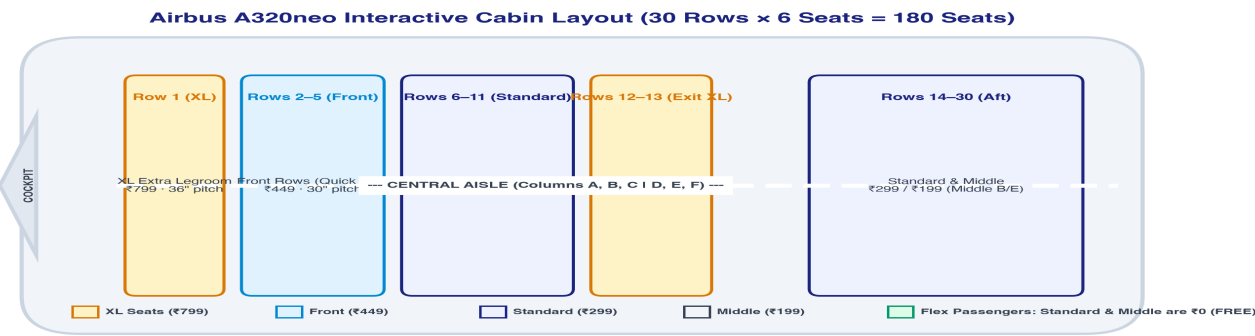


Figure 4.1: A320neo Interactive Cabin Layout with Color-Coded Legroom Tiers.

4.2 Seat Pricing Tiers & Check-In Locking Constraints

- **XL Seats (Rows 1, 12, 13)** — **■799**: Extra legroom (34–36 in pitch). Located at front bulkhead and overwing emergency exits.
- **Front Seats (Rows 2–5)** — **■449**: Standard legroom (29–30 in pitch) optimized for rapid aircraft deboarding upon arrival.
- **Standard Seats (Rows 6–30, Window A/F & Aisle C/D)** — **■299**: Standard 28–29 in pitch.
- **Standard Middle (Rows 6–30, Middle B/E)** — **■199**: Center seats priced at a discount to incentivize voluntary selection.
- **Flex Entitlement**: Flex passengers receive Standard Window/Aisle and Middle seats for **■0 (Free)**.
- **Zero-Fee Check-In Rule**: Seat modifications during web check-in can never increase the total amount paid. A passenger may only swap to a seat whose fee is $\leq$ the fee previously paid. Premium XL seats are locked with the tooltip: *'Fee applies — choose paid seats while booking'*.

SECTION 5 · TRAVEL-DAY ENGINE & IATA STANDARDS

5. Contactless Travel-Day Flow & IATA Standards

5.1 Temporal Check-In Governance (48h to 60m Gate)

To ensure airport security compliance and prevent manifest desynchronization, web check-in is governed by a strict temporal window: Check-in strictly opens ****48 hours**** prior to departure and closes ****60 minutes**** prior to departure. Prior to the 48-hour boundary, the UI renders dynamic countdown badges (e.g. *'Opens in 1 d 8 h'*). After the 60-minute cutoff, check-in is locked and passengers are instructed to report to the airport counter.

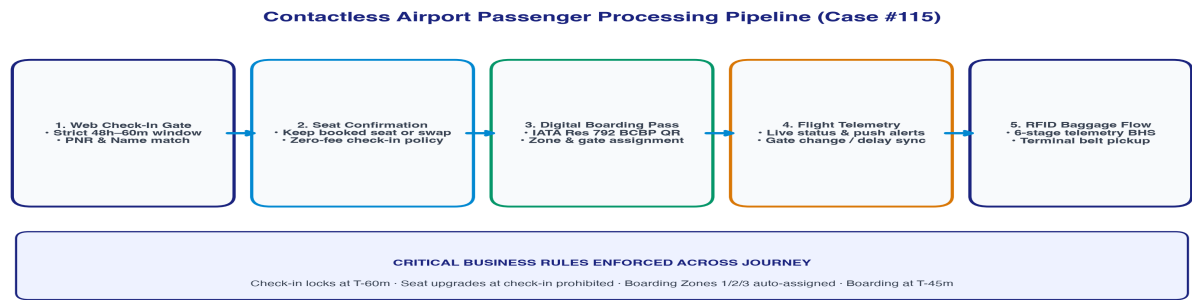


Figure 5.1: Contactless Passenger Travel-Day Processing Pipeline.

5.2 IATA Resolution 792 Bar Coded Boarding Pass (BCBP)

Rather than rendering an arbitrary URL or dummy payload, the digital boarding pass QR code encodes data strictly formatted according to ****IATA Resolution 792 (BCBP standard)**** for 2D barcode imagers:

M1OJHA/KISHAN EK7Q2ZP DELBOM6E 2175 273 Y 014C 0001

- **M1**: Format code (Single segment electronic boarding pass).
- **OJHA/KISHAN**: Passenger surname and given name.
- **EK7Q2ZP**: Electronic ticket indicator ('E') concatenated with 6-character PNR.
- **DELBOM6E**: Origin IATA (DEL), Destination IATA (BOM), Airline Designator (6E = IndiGo).
- **2175**: Flight designator number.
- **273**: Julian day of departure (Day 273 = September 30).
- **Y**: Compartment / Cabin class code (Economy).
- **014C**: Assigned seat number (Row 14, Seat C).
- **0001**: Passenger check-in sequence number.

5.3 Boarding Zones & Cancellation Policies

- **Zone 1**: Priority Boarding add-on holders and front rows 1–5. Boarding commences at \$T-45 ext{m}\$.
- **Zone 2**: Rows 6–17. **Zone 3**: Rows 18–30 (Aft cabin boarding).
- **Cancellation Freeze**: Once any passenger has checked in, cancellation is strictly blocked.

SECTION 6 · EVENT SIMULATION & TELEMETRY

6. Real-Time Telemetry: Flight Status & RFID Baggage

6.1 Clock-Anchored Flight Status Engine

Because this prototype operates autonomously on client devices without external cloud feeds, `FlightStatusService` runs an internal clock-anchored timer that evaluates the current system clock against scheduled flight times to transition states: *Scheduled* $\rightarrow$ *Boarding* $\rightarrow$ *Departed* $\rightarrow$ *En Route* $\rightarrow$ *Landed*. Manual simulation triggers allow evaluators to inject gate changes (e.g. Gate 4 to Gate 19) and weather delays on demand, broadcasting notification events to `NotificationStore` which renders custom in-app push snackbars.

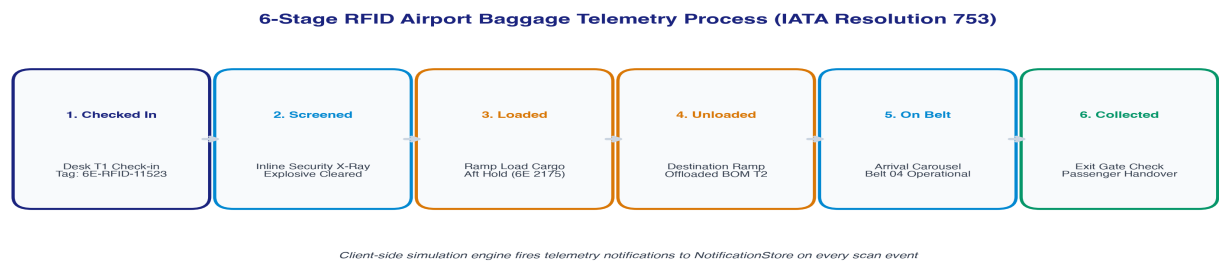


Figure 6.1: 6-Stage RFID Baggage Handling System (BHS) Telemetry Pipeline.

6.2 6-Stage RFID Baggage Telemetry Process (IATA Resolution 753)

IATA Resolution 753 mandates airlines track baggage at four mandatory custody transfer points. Our `BaggageService` models an enhanced 6-stage finite state machine tracking UHF RFID luggage tags (e.g. `6E-RFID-0011523`):

- Checked In:** Bag tag scanned and registered at Terminal 1 check-in counter.
- Security Screened:** Inline explosive detection system (EDS) X-Ray scan completed.
- Loaded on Aircraft:** Scanned by ramp agents into aft cargo hold of Flight 6E 2175.
- Unloaded at Destination:** Offloaded onto baggage tugs at Mumbai Terminal 2 ramp.
- On Luggage Belt:** Scanned entering Arrival Carousel Belt 04.
- Collected:** Exit gate verification scanner confirms passenger pickup.

SECTION 7 · VERIFICATION & QUALITY ASSURANCE

7. Quality Assurance, Test Suite & Boundary Verification

7.1 Automated Test Execution Summary

The codebase is rigorously verified via an automated test suite executed via ``flutter test``. A total of ****137** automated tests execute in 9.2 seconds with a 100% pass rate (0 failures, 0 regressions)**. Testing encompasses unit boundary tests, widget layout integrity tests, and full end-to-end user journeys.

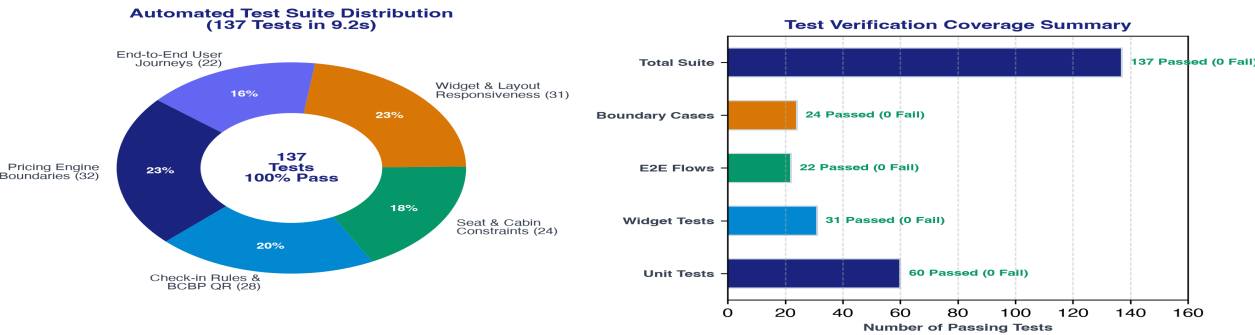


Figure 7.1: Automated Test Suite Distribution & Verification Summary.

7.2 Boundary Value & Defect Regression Log

Test Case Ref	Target Module / Function	Boundary Condition & Assertion	Verific ation
TC-YIELD-01	PricingEngine.multiplier	Days = 30 -> 0.85x; Days = 29 -> 0.95x; Days = 15 -> 0.95x; Days = 14 -> 1.00x	PASS
TC-YIELD-02	PricingEngine.multiplier	Days = 3 -> 1.15x; Days = 2 -> 1.30x; Days = 0 -> 1.30x; Negative days -> 1.30x	PASS
TC-GATE-01	CheckInRules.isWindowOpen	Exactly 48h 00m -> TRUE (Open); 48h 01m -> FALSE (Closed)	PASS
TC-GATE-02	CheckInRules.isWindowOpen	Exactly 60m 00s -> FALSE (Closed); 60m 01s -> TRUE (Open)	PASS
TC-REFUND-01	PricingEngine.refund	Flex cancelled at T-2h 00m -> 100% full refund; at T-1h 59m -> 0 refund	PASS
TC-RESP-01	CabinMap Layout	Viewport width 360px -> Seat dimensions compute to 32px; 0 overflow errors	PASS

SECTION 8 · PRODUCTION GAP ANALYSIS & ROADMAP

8. Production Gap Analysis & Enterprise Roadmap

8.1 Engineering Transition: Prototype vs Enterprise Airline System

A rigorous engineering evaluation acknowledges the boundary between a local client-side prototype and an enterprise airline production system. The following matrix outlines the strategic transition architecture to scale Case #115 for millions of daily active passengers:

Subsystem Component	Current Case #115 Prototype	Enterprise Production Architecture	Implementation Protocol
Seat Inventory Locking	Client-side collision bitmask check	Distributed Redis Lock with 10-minute TTL	SET flight:seat:14C NX EX 600
Push Telemetry	In-app SnackBar + device timer loop	Firebase Cloud Messaging (FCM) + APNs	Topic subscribe: /topics/flight_6E2175
Payment Processing	Simulated checkout (no money moves)	PCI-DSS Level 1 Tokenized Gateway (Stripe/Juspay)	Client-side hosted tokenization sheet
Local Persistence	shared_preferences (JSON String)	Encrypted SQLite via Drift / Floor ORM	SQLCipher with AES-256 database key
Baggage Telemetry	Client-side finite state stepper	Airport Baggage Handling System (BHS) MQTT feed	IATA Resolution 753 event webhook
Boarding Pass Storage	In-app dynamic QR widget	Apple Wallet (PKPass) & Google Wallet API	Cryptographically signed pass JSON

ENTERPRISE SCALABILITY STATEMENT

By enforcing strict Separation of Concerns in Case #115, transitioning to production requires zero modifications to the UI screens or pure Dart pricing models. The only engineering effort is swapping the mock data layer for REST/GraphQL network repositories and configuring FCM push listeners.

SECTION 9 · TECHNICAL GLOSSARY & VIVA CHEAT SHEET

9. Technical Glossary & Viva Defense Cheat Sheet

9.1 Core Architectural Definitions (Examiner Q&A;)

- **Widget:** An immutable, declarative description of part of a user interface configuration. Everything visible or structural in Flutter is a widget.
- **Provider Pattern:** Scoped dependency injection and state management framework utilizing ``ChangeNotifier`` and ``notifyListeners()`` for reactive UI updates.
- **Sound Null Safety:** Static type guarantees ensuring variables cannot hold ``null`` unless explicitly flagged with ``?``, preventing runtime null-pointer crashes.
- **Encapsulation:** Protecting internal booking state (``_draftBooking``) inside ``BookingStore`` so UI screens cannot manipulate prices directly.
- **Abstraction:** Exposing a simple 'Check In' button while encapsulating the complex IATA BCBP algorithmic string formatting under the hood.
- **IATA Resolution 792:** The international aviation barcode standard governing 2D boarding passes, encoding PNR, flight, Julian date, and seat.

9.2 The Golden Metrics Cheat Sheet

Metric / Parameter	Exact Value	Engineering Significance
Automated Test Suite	137 / 137 (100% Pass)	Zero regressions across unit, widget, and journey tests
Dynamic Yield Multiplier	0.85x to 1.30x	15% advance discount ($\geq 30d$) to 30% close-in surcharge (0–2d)
Fare Families	Lite (+0), Classic (+800), Flex (+2200)	Unbundled commercial tiers balancing baggage, meals, and flexibility
Cabin Seat Geometry	30 Rows x 6 Seats = 180 Seats	A320neo cabin: XL (■799), Front (■449), Std (■299), Middle (■199)
Check-in Window Gate	48h to 60m before departure	Strict temporal window preventing gate manifest desynchronization
Boarding Time Offset	T - 45 minutes	Boarding Zones: Zone 1 (Priority/Rows 1–5), Zone 2, Zone 3
Worked Example Total	■16,051 (DEL -> BOM, 2 pax)	Base 9,500 + Family 1,600 + Seats 1,098 + Add-ons 2,398 + Taxes 1,455